Report: Optimizing NYC Taxi Operations__Anil_Singal

Include your visualizations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

Objective:
To load the NYC taxi data from multiple files and combine them into one dataset for analysis.

Methodology:
Since the dataset is very large, a small sample was taken from each file instead of loading the full data. Around **0.007 fraction** of data was sampled from every monthly file. These sampled datasets were then combined into a single DataFrame. This helped reduce memory usage while still keeping the overall data patterns.

Result:
A combined dataset was created with all important trip details like pickup time, dropoff time, distance, fare, and location.

Conclusion:
Sampling made it easier to work with large data while still maintaining useful information for analysis.

Sample data after loading:

VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	RatecodeID	store
0	2	1/1/2023 0:07	1/1/2023 0:23	1	7.74	
1	2	1/1/2023 0:16	1/1/2023 0:21	2	1.24	
2	2	1/1/2023 0:14	1/1/2023 0:24	3	1.44	
3	2	1/1/2023 0:24	1/1/2023 0:29	1	0.54	
4	2	1/1/2023 0:43	1/1/2023 1:01	NaN	19.24	

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

The dataset index was reset to ensure proper row numbering after combining and sampling the data, making the dataset ready for further cleaning and analysis.

2.1.2. Combine the two airport_fee columns

The dataset contained two similar columns **airport_fee** and **Airport_fee**, which represented the same information but had missing values in different rows. First, the values and missing data in both columns were checked using summary statistics and null value counts. Then, the missing values in the **airport_fee** column were filled using the values from the **Airport_fee** column. After combining the information into a single column, the **Airport_fee** column was dropped to remove duplication and keep the dataset clean and consistent.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

First, I checked the percentage of missing values in each column to understand how much data was missing and which columns needed cleaning. This helped in deciding how to handle missing values for different columns.

2.2.2. Handling missing values in passenger_count

The passenger_count column had some missing values. Since passenger count is a numeric field, I filled the missing values using the median passenger count so that the data distribution would not be affected by outliers.

2.2.3. Handle missing values in RatecodeID

RatecodeID is a categorical column, so the missing values were filled using the most frequent value (mode). This is because most trips belong to the standard rate type.

2.2.4. Impute NaN in congestion_surcharge

The missing values in congestion_surcharge were replaced with 0, assuming that if the surcharge value is missing, then no congestion surcharge was applied for that trip.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

I checked the **payment_type, trip_distance, and tip_amount** columns to identify any unusual or incorrect values. For payment_type, I looked for invalid categories like 0 which are not defined in the data dictionary. For trip_distance, I checked for very high values and also cases where distance was zero but the trip looked unrealistic. For tip_amount, I checked for negative values and extremely high tips which could be outliers or data entry errors.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

First, the variables in the dataset were classified into categorical and numerical variables. Categorical variables included fields such as VendorID, payment_type, RatecodeID, PULocationID, and DOLocationID because they represent categories or IDs. Numerical variables included trip_distance, fare_amount, tip_amount, total_amount, passenger_count, and surcharges because they represent numeric values that can be measured and analyzed statistically. This classification helped in choosing the correct analysis and visualisation methods for different variables.

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

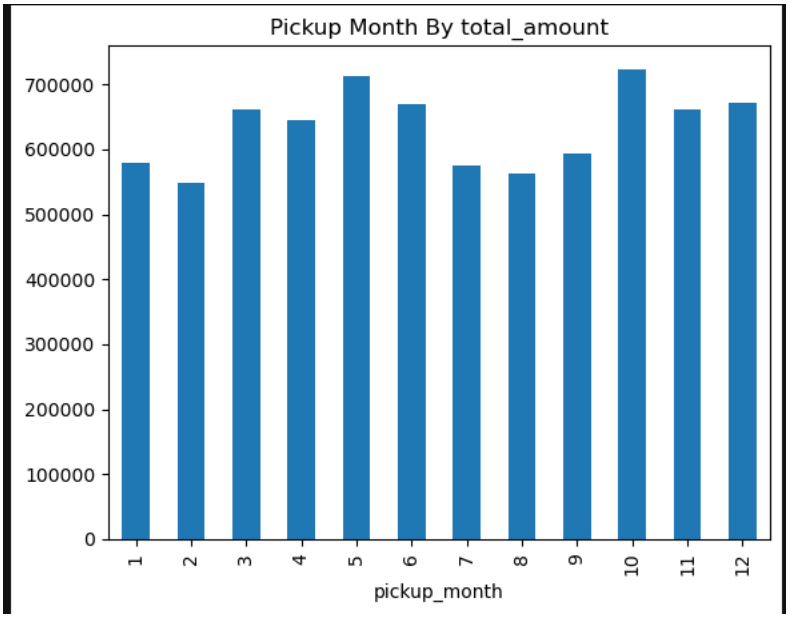
First I have converted pickup date in proper format, then in Pickup datetime was used to extract hour, day of week, and month. Then the number of trips was calculated for each hour, day, and month to understand demand patterns. From the analysis, it was observed that taxi demand is higher during peak office hours and evening hours, lower during late night and early morning hours, higher on weekends compared to weekdays, and varies across months depending on season and holidays. This analysis helped in understanding demand trends over time.

3.1.3. Filter out the zero/negative values in fares, distance and tips

Trips with zero or negative values in fares amount, trip distance, and tip amount were filtered out because these values are unrealistic and indicate data entry errors or cancelled trips. Removing these incorrect records helped in improving the accuracy of further analysis like fare analysis and distance analysis.

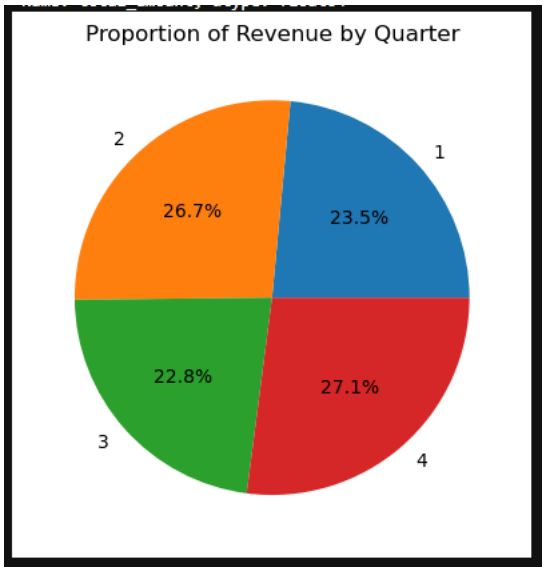
3.1.4. Analyse the monthly revenue trends

Monthly revenue was calculated by grouping the data by month and summing the total amount. The monthly revenue trend showed how taxi business revenue changes across different months. This helped in identifying high revenue months and low revenue months, which may be affected by seasonality, holidays, tourism, or weather conditions.



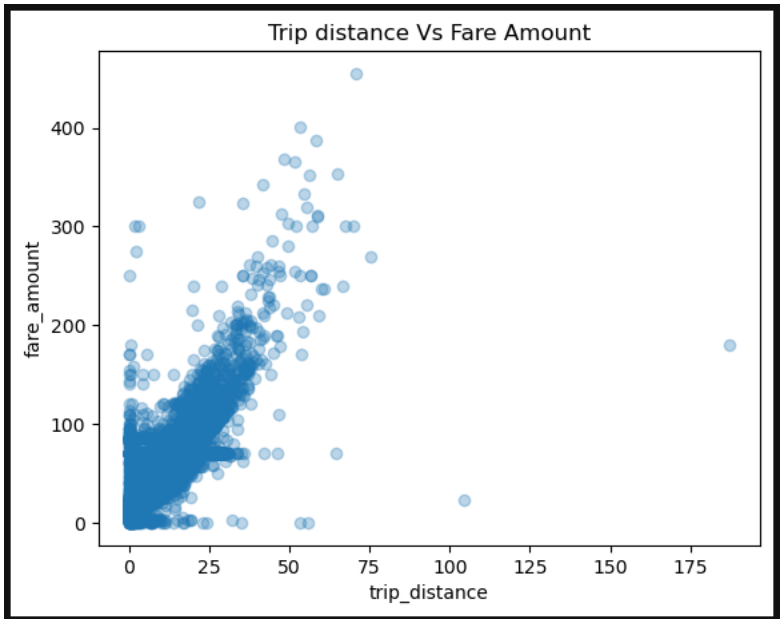
3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

The total revenue for each quarter was calculated and then divided by the total yearly revenue to find the proportion contributed by each quarter. This analysis helped in understanding which quarter contributes the most revenue and which quarter has lower revenue contribution in the year.



3.1.6. Analyse and visualise the relationship between distance and fare amount

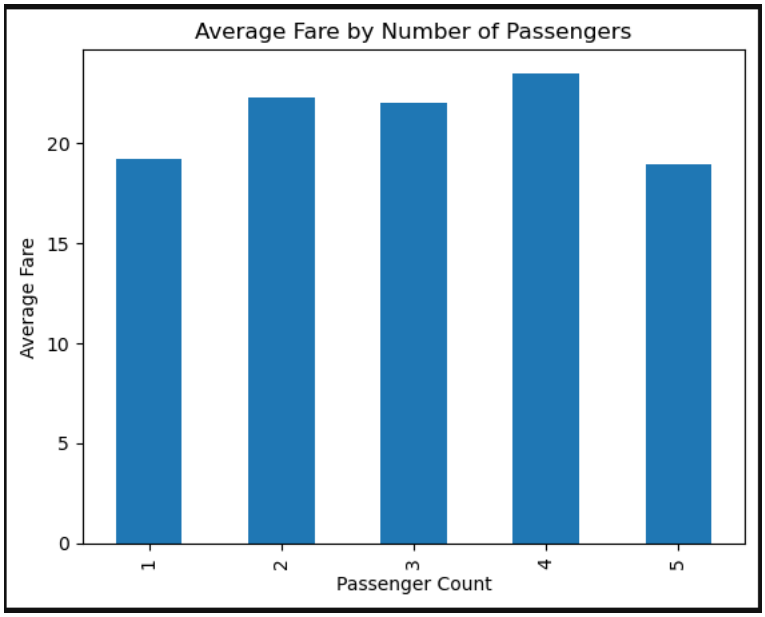
A scatter plot was used to analyze the relationship between trip distance and fare amount. The analysis showed that fare amount generally increases as trip distance increases, indicating a positive relationship between distance and fare. However, some variations were observed due to surcharges, traffic time, and extra charges.



3.1.7. Analyse the relationship between fare/tips and trips/passengers

The relationship between fare, tips, number of trips, and passenger count was analyzed using groupby and average calculations. It was observed

that trips with higher passenger counts sometimes have higher total fare and tips. Also, more trips generally result in higher overall revenue and tips collected.

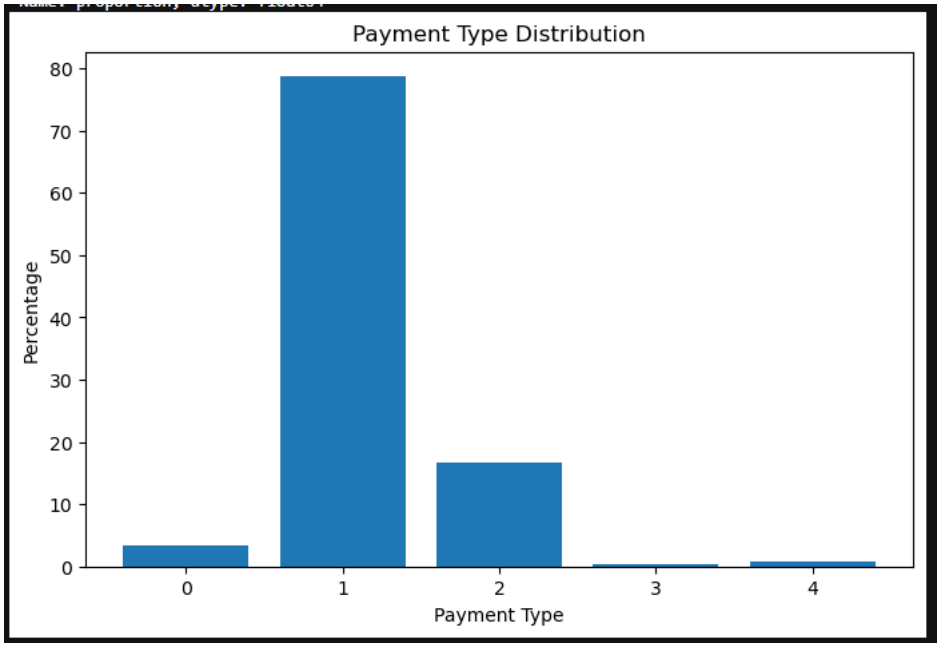


3.1.8. Analyse the distribution of different payment types

The distribution of payment types was analyzed by counting the number of trips for each payment type. The analysis showed that most customers prefer card payments, followed by cash payments, while other payment types are used less frequently.

```
Total By Payment_type
1    207214
2     44061
0      8828
4      1907
3      1216

Payment Percentage By payment_type
1    78.720947
2    16.738848
0     3.353772
4     0.724473
3     0.461960
```



3.1.9 Load the taxi zones shapefile and display it

The taxi zone shapefile was loaded using GeoPandas and displayed to visualize the geographical zones of New York City.

```
# Read the shapefile using geopandas
zones = gpd.read_file(r"D:\IIITB\NYC Taxi EDA Project\Datasets and Dictionary-NYC\Datasets and Dictionary\taxi_zones\taxi_zones.shp") #
zones.head()
```

	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry
0	1	0.116357	0.000782	Newark Airport	1	EWR	POLYGON ((933100.918 192536.086, 933091.011 19...
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...

3.1.10 Merge the zone data with trips data

For spatial analysis, the trip dataset was merged with the taxi zone dataset using a left join on PULocationID and LocationID. This allowed us to map each trip to its corresponding pickup zone and perform location-based analysis of trip patterns, demand, and revenue distribution.

3.1.11 Find the number of trips for each zone/location ID

The number of trips for each zone was calculated by grouping the trips data using the pickup location ID and counting the number of

trips for each location. This helped in identifying which zones have the highest and lowest taxi activity.

Below here sample output:

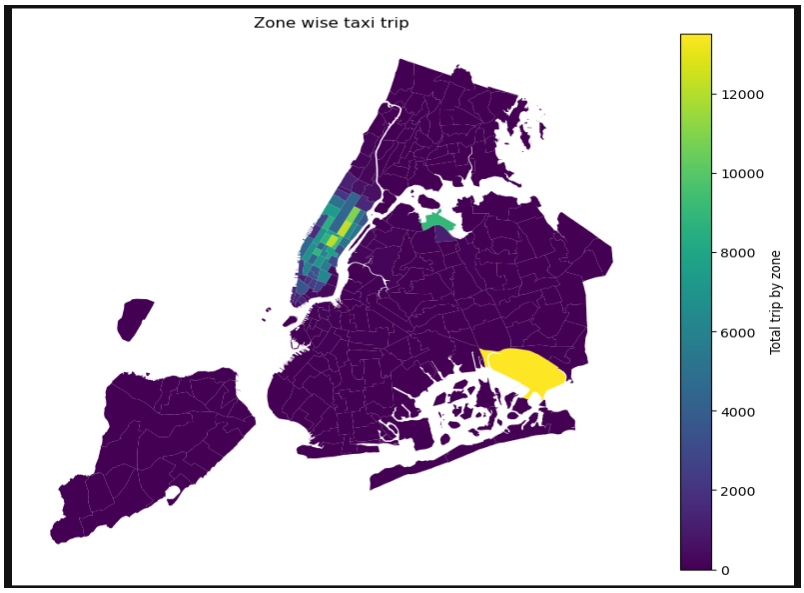
	PULocationID	total_trips
0	1	37
1	3	9
2	4	308
3	5	1
4	6	3
5	7	136
6	9	3
7	10	81
8	11	5
9	12	110

3.1.12 Add the number of trips for each zone to the zones dataframe

After calculating the number of trips per zone, the results were merged back into the zones GeoDataFrame. This added a new column showing the total number of trips for each zone, which was useful for visualization and spatial analysis.

	OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	borough	geometry	PULocationID	total_trips
0	1	0.116357	0.000782	Newark Airport	1	EWR	POLYGON ((933100.918 192536.086, 933091.011 19...	1.0	37.0
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...	NaN	NaN
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...	3.0	9.0

3.1.13 Plot a map of the zones showing number of trips



3.1.14 Conclude with results

From the spatial analysis, it was observed that taxi demand is not evenly distributed across all zones. A few zones contribute a large portion of the total trips, especially commercial areas, busy city centers, residential pickup areas during peak hours, and airport zones. These zones showed consistently high pickup activity, while some outer zones and less populated areas had very low taxi usage.

The zone-wise trip distribution map helped clearly identify high-demand zones and low-demand zones. High-demand zones indicate areas where more taxis should be positioned to reduce passenger waiting time and increase trip frequency. Low-demand zones indicate areas where taxis may experience more idle time, so dispatching should be optimized to avoid unnecessary waiting in these areas.

This spatial analysis is very useful for operational planning. Taxi companies can use this information to strategically position taxis in high-demand zones, improve dispatching decisions, reduce empty trips, and increase overall fleet utilization. It can also help in demand forecasting and surge pricing strategies in high-demand locations. Overall, zone-based analysis helps improve revenue, reduce operational inefficiencies, and provide better service to customers.

3.2 Detailed EDA: Insights and Strategies

3.2.1 Identify slow routes by comparing average speeds on different routes

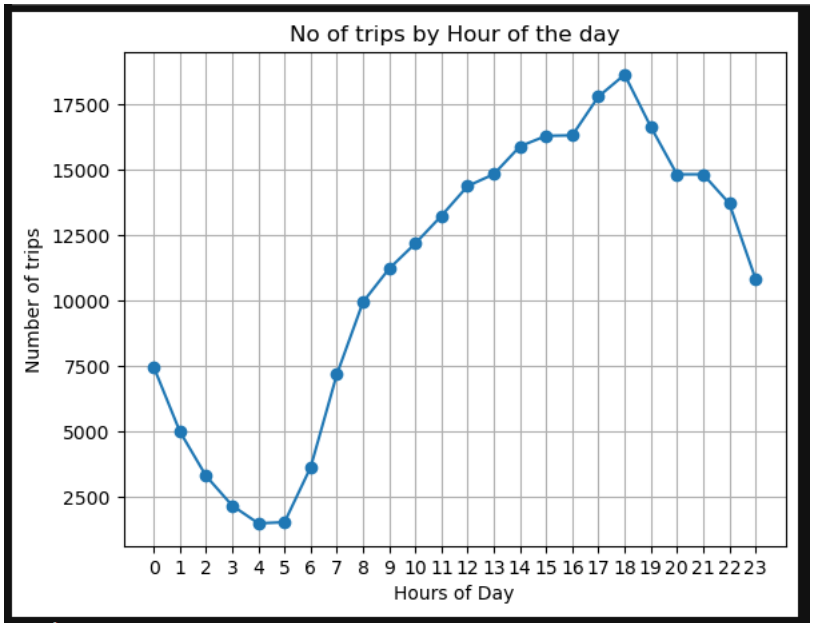
To identify slow routes, the average speed for each route was calculated using trip distance and trip duration. Then the average

speed was compared across different pickup and dropoff zone pairs. Routes with very low average speeds were identified as slow routes, which may be affected by traffic congestion, signals, or busy city areas. Identifying slow routes helps in improving route planning and reducing trip time.

	PULocationID	DOLocationID	pickup_hour	trip_distance	trip_duration	avg_speed
•	88	144	0	1.780000	1425.466667	0.074923
	164	263	1	3.488000	296.023333	0.706971
	229	137	2	1.940000	1434.250000	0.081157
	148	148	3	0.756429	105.221429	0.431335
	230	51	4	16.960000	1417.666667	0.717799

3.2.2 Calculate the hourly number of trips and identify the busy hours

The number of trips was calculated for each hour using pickup time data. After grouping trips by hour, it was observed that taxi demand is higher during morning office hours, evening peak hours, and late-night hours on weekends. These hours were identified as busy hours because they had the highest number of trips.



3.2.3 Scale up the number of trips from above to find the actual number of trips

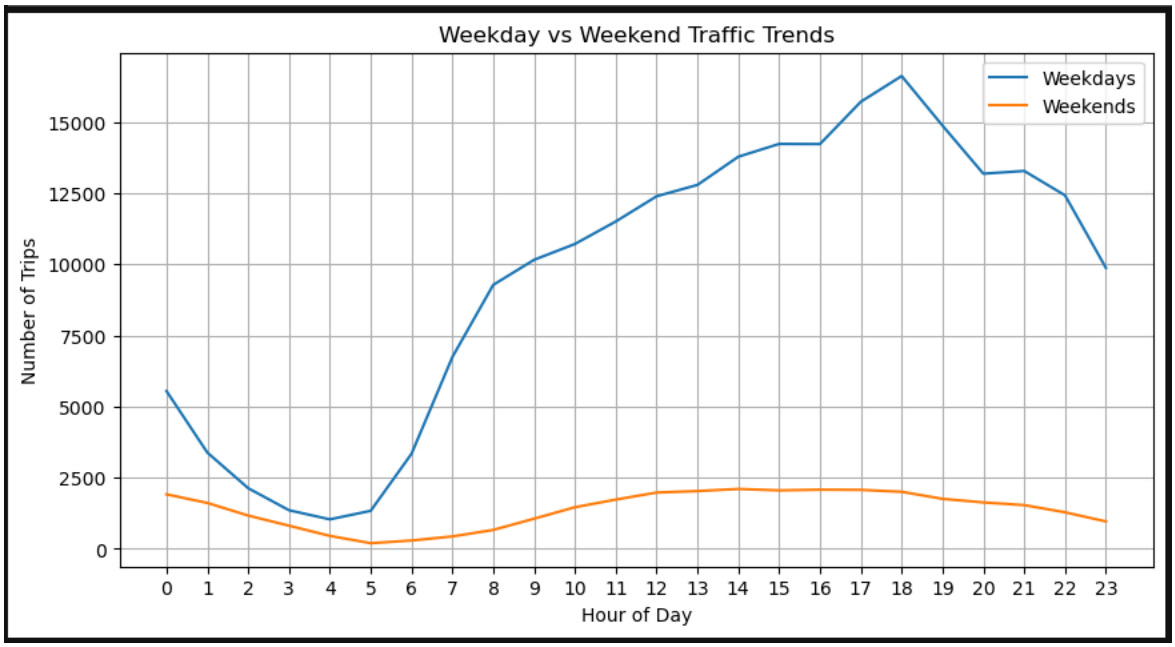
Since only a sample of the dataset was used for analysis, the hourly trip counts were scaled up using the sampling fraction to

estimate the actual number of trips. This helped in understanding the real demand pattern across hours instead of just the sample data.

	pickup_hour	sample_trips	actual_trips
18	18	18624	2660571
17	17	17789	2541285
19	19	16634	2376285
16	16	16307	2329571
15	15	16288	2326857

3.2.4 Compare hourly traffic on weekdays and weekends

Trips were grouped by hour separately for weekdays and weekends and then compared. It was observed that weekday traffic is higher during office commute hours, while weekend traffic is higher during afternoon and night hours. Weekend nights showed significantly higher taxi usage compared to weekdays.



3.2.5 Identify the top 10 zones with high hourly pickups and drops

The number of pickups and dropoffs was calculated for each zone and sorted to find the top 10 zones with the highest activity. These zones represent the busiest areas where taxi demand is consistently high. These areas are important for taxi positioning and dispatch planning.

Top 10 Pickups:

	zone	borough	total_pickups
117	JFK Airport	Queens	13522
214	Upper East Side South	Manhattan	12228
145	Midtown Center	Manhattan	12177
213	Upper East Side North	Manhattan	10961
146	Midtown East	Manhattan	9336
123	LaGuardia Airport	Queens	8944
166	Penn Station/Madison Sq West	Manhattan	8795
207	Times Sq/Theatre District	Manhattan	8674
127	Lincoln Square East	Manhattan	8573
154	Murray Hill	Manhattan	7714

Top 10 Drop-offs:

	zone	borough	total_dropoffs
226	Upper East Side North	Manhattan	11672
227	Upper East Side South	Manhattan	10912
154	Midtown Center	Manhattan	10201
220	Times Sq/Theatre District	Manhattan	7967
163	Murray Hill	Manhattan	7772
155	Midtown East	Manhattan	7437
135	Lincoln Square East	Manhattan	7400
229	Upper West Side South	Manhattan	7303
134	Lenox Hill West	Manhattan	6766
67	East Chelsea	Manhattan	6724

3.2.6 Find the ratio of pickups and dropoffs in each zone

The pickup to dropoff ratio was calculated for each zone by dividing total pickups by total dropoffs. Zones with a high pickup-to-dropoff ratio indicate areas where more passengers start trips, while zones with a low ratio indicate areas where more trips end. This helps in understanding taxi movement and repositioning taxis efficiently.

Pickups ratio:

	zone	borough	top_10_pickup_ratio
117	JFK Airport	Queens	0.051370
214	Upper East Side South	Manhattan	0.046454
145	Midtown Center	Manhattan	0.046261
213	Upper East Side North	Manhattan	0.041641
146	Midtown East	Manhattan	0.035468
123	LaGuardia Airport	Queens	0.033978
166	Penn Station/Madison Sq West	Manhattan	0.033412
207	Times Sq/Theatre District	Manhattan	0.032953
127	Lincoln Square East	Manhattan	0.032569
154	Murray Hill	Manhattan	0.029306

Drop-offs ratios:

	zone	borough	top_10_dropoff_ratio
226	Upper East Side North	Manhattan	0.044342
227	Upper East Side South	Manhattan	0.041455
154	Midtown Center	Manhattan	0.038754
220	Times Sq/Theatre District	Manhattan	0.030267
163	Murray Hill	Manhattan	0.029526
155	Midtown East	Manhattan	0.028253
135	Lincoln Square East	Manhattan	0.028113
229	Upper West Side South	Manhattan	0.027744
134	Lenox Hill West	Manhattan	0.025704
67	East Chelsea	Manhattan	0.025545

3.2.7 Identify the top zones with high traffic during night hours

Trips between 11 PM and 5 AM were filtered and then pickup and dropoff counts were calculated for each zone. The top zones during night hours were identified based on the highest number of

trips. These zones usually include airports, entertainment areas, and late-night activity areas.

	zone	borough	total_pickups
53	East Village	Manhattan	2294
84	JFK Airport	Queens	2032
173	West Village	Manhattan	1864
30	Clinton East	Manhattan	1473
101	Lower East Side	Manhattan	1432
157	Times Sq/Theatre District	Manhattan	1232
77	Greenwich Village South	Manhattan	1219
126	Penn Station/Madison Sq West	Manhattan	984
113	Midtown South	Manhattan	900
92	LaGuardia Airport	Queens	881

	zone	borough	total_dropoffs
72	East Village	Manhattan	1222
43	Clinton East	Manhattan	1029
154	Murray Hill	Manhattan	893
61	East Chelsea	Manhattan	866
95	Gramercy	Manhattan	849
127	Lenox Hill West	Manhattan	780
240	Yorkville West	Manhattan	764
226	West Village	Manhattan	686
82	Flatiron	Manhattan	665
134	Lower East Side	Manhattan	660

3.2.8 Find the revenue share for nighttime and daytime hours

Total revenue was calculated separately for daytime and nighttime trips. Then the revenue share was calculated as a percentage of total revenue. This analysis helped in understanding whether more revenue comes from daytime trips or nighttime trips.

```
3.3 Night Revenue Share: 12.351686155764021
3.4 Day Revenue Share: 87.64831384423599
```

3.2.9 For the different passenger counts, find the average fare per mile per passenger

Trips were grouped by passenger count, and the average fare per mile per passenger was calculated. This analysis helped in understanding how pricing and revenue vary based on the number of passengers. It also helped identify whether trips with more

passengers generate more revenue per mile compared to single-passenger trips.

	passenger_count	fare_per_passenger
0	1	10.779463
1	2	6.185374
2	3	4.332130
3	4	5.182434
4	5	1.565300

3.2.10 Find the average fare per mile by hours of the day and by days of the week

The fare per mile was calculated by dividing fare amount by trip distance. Then the average fare per mile was calculated for each hour of the day and for each day of the week. From the analysis, it was observed that fare per mile is higher during peak traffic hours such as morning and evening because trips take longer due to traffic congestion. Fare per mile also varies across days, with slightly higher values on weekends and busy days due to higher demand and traffic conditions.

	pickup_dayofweek	fare_per_mile
4	Thursday	12.764576
3	Sunday	11.837198
6	Wednesday	11.581904
2	Saturday	11.131689
5	Tuesday	10.943397
1	Monday	10.232056
0	Friday	9.945052

3.2.11 Analyse the average fare per mile for the different vendors

The average fare per mile was calculated separately for each vendor by grouping the data by VendorID. The analysis showed that fare per mile is slightly different across vendors, which may be due to different pricing structures, trip types, or areas where the vendors operate more frequently. This comparison helps in understanding pricing differences between vendors.

	VendorID	fare_per_mile
0	1	8.010863
1	2	12.402533
2	6	5.050381

3.2.12 Compare the fare rates of different vendors in a distance-tiered fashion

Trips were divided into different distance categories such as short trips (0–2 miles), medium trips (2–5 miles), and long trips (more than 5 miles). Then the average fare per mile was calculated for each vendor within each distance category. The analysis showed that fare per mile is higher for short trips and decreases as distance increases. This is expected because short trips include base fare and minimum charges, while longer trips have more distance-based pricing.

	VendorID	distance_tier	fare_per_mile
0	1	0-2 miles	9.749953
1	1	2-5 miles	6.385617
2	1	>5 miles	4.411753
3	2	0-2 miles	18.087303
4	2	2-5 miles	6.533473

3.2.13 Analyse the tip percentages

Tip percentage was calculated by dividing tip amount by fare amount. Then the tip percentage was analysed across different distances, passenger counts, and pickup hours. The analysis showed that tip percentage is generally higher for medium and long trips compared to very short trips. Tips were also slightly higher during evening and night hours when demand is higher.

distance_tier	tip_percentage
0-2 miles	22.076652
2-5 miles	18.583716
>5 miles	16.489020

3.2.14 Analyse the trends in passenger count

Passenger count was analyzed across different hours and days to understand travel patterns. It was observed that most trips have 1 or 2 passengers, while trips with higher passenger counts are less frequent. Passenger counts are slightly higher during weekends and evening hours when people travel in groups.

pickup_hour	passenger_count
0	1.347272
1	1.377978
2	1.360715
3	1.329178
4	1.260753
pickup_dayofweek	passenger_count
Friday	1.333359
Monday	1.289627
Saturday	1.401632
Sunday	1.392445
Thursday	1.275766
Tuesday	1.265744
Wednesday	1.262534

3.2.15 Analyse the variation of passenger counts across zones

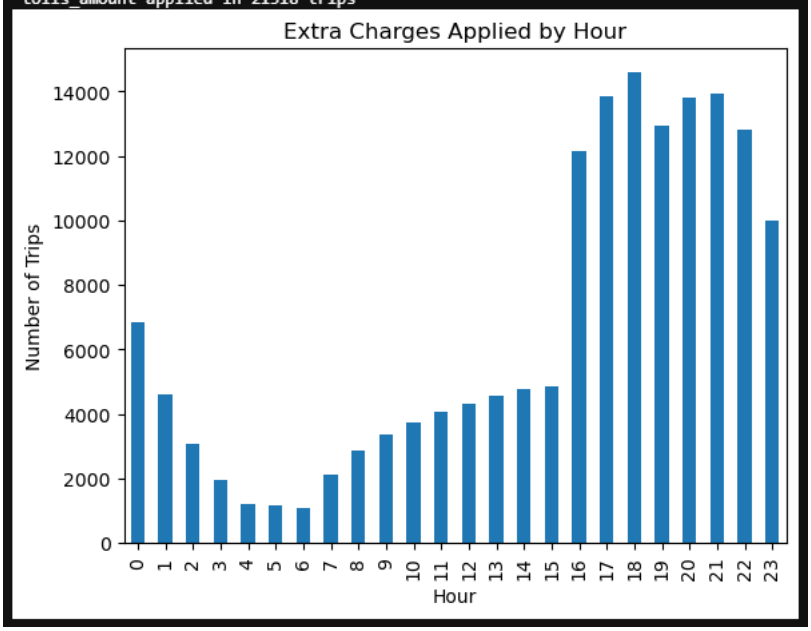
The average passenger count was calculated for each pickup zone and then merged with the zones dataset. This helped in identifying zones where trips usually have more passengers. Zones such as airports, tourist areas, and residential areas showed higher average passenger counts compared to business districts where single passenger trips are more common.

	zone	borough	passenger_count
3	Arrochar/Fort Wadsworth	Staten Island	2.000000
58	Dyker Heights	Brooklyn	2.000000
136	Marine Park/Floyd Bennett Field	Brooklyn	2.000000
41	Claremont/Bathgate	Bronx	1.727273
173	Red Hook	Brooklyn	1.680000

3.2.16 Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

Extra charges such as airport fees, congestion surcharge, and other surcharges were analyzed across zones and pickup hours. It was observed that extra charges are applied more frequently in airport zones, busy city areas, and during peak traffic hours. Congestion surcharge was more common in busy zones and peak hours, while airport fees were applied mainly for airport pickup and dropoff trips.

```
extra applied in 158568 trips
mta_tax applied in 260794 trips
improvement_surcharge applied in 263112 trips
congestion_surcharge applied in 243690 trips
airport_fee applied in 22441 trips
tolls_amount applied in 21318 trips
```



Extra charges By Pickup Zone:

	PULocationID	extra_trips	LocationID	zone	borough
81	138	8838	138.0	LaGuardia Airport	Queens
97	161	7812	161.0	Midtown Center	Manhattan
146	237	6975	237.0	Upper East Side South	Manhattan
76	132	6707	132.0	JFK Airport	Queens
140	230	5746	230.0	Times Sq/Theatre District	Manhattan
145	236	5743	236.0	Upper East Side North	Manhattan
98	162	5732	162.0	Midtown East	Manhattan
84	142	5270	142.0	Lincoln Square East	Manhattan
112	186	5152	186.0	Penn Station/Madison Sq West	Manhattan
99	163	4671	163.0	Midtown North	Manhattan

Extra charges By Dropoff zone:

	DOLocationID	extra_trips	LocationID	zone	borough
225	236	6733	236.0	Upper East Side North	Manhattan
226	237	5952	237.0	Upper East Side South	Manhattan
153	161	5040	161.0	Midtown Center	Manhattan
162	170	4712	170.0	Murray Hill	Manhattan
219	230	4603	230.0	Times Sq/Theatre District	Manhattan
228	239	4496	239.0	Upper West Side South	Manhattan
134	142	4392	142.0	Lincoln Square East	Manhattan
133	141	4359	141.0	Lenox Hill West	Manhattan
45	48	4191	48.0	Clinton East	Manhattan
154	162	3994	162.0	Midtown East	Manhattan

4 Conclusions

4.2 Final Insights and Recommendations

4.1.1 Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

Based on the analysis of trip demand, traffic speeds, trip durations, and zone-wise trip distribution, routing and dispatching can be improved by allocating taxis dynamically based on demand patterns and traffic conditions. Taxis should avoid slow routes during peak traffic hours and instead use faster alternate routes to reduce trip time. Dispatch systems should assign taxis based on nearest high-demand areas rather than random allocation. This will reduce idle time, reduce empty trips, increase the number of trips per taxi, and improve overall operational efficiency.

4.1.2 Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analyzing trip trends across time, days and months.

From the zone and time-based analysis, it was observed that demand varies across different zones and different times such as peak hours, weekends, and night hours. Taxis should be strategically positioned in high-demand pickup zones during peak hours, residential areas during morning hours, commercial areas during evening hours, and airport or entertainment zones during night hours. During low-demand periods, taxis should be repositioned to nearby high-demand zones instead of waiting in low-demand areas. This strategy will help reduce passenger waiting time, increase taxi utilization, and improve revenue generation.

4.1.3 Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Based on fare per mile, trip distance, demand patterns, and time-based revenue analysis, pricing strategies can be adjusted using a data-driven approach. Higher pricing can be applied during peak demand hours, busy zones, and high-traffic periods where demand is high. For short trips, minimum fare pricing can be optimized since fare per mile is higher for shorter distances. Discounts or promotional pricing can be introduced during low-demand hours to increase trip volume. Nighttime and airport trips can include additional charges since demand is consistent during those periods. These pricing adjustments can help maximize revenue while still remaining competitive with other vendors.